35th Annual University of Alabama System Applied Mathematics Meeting

Saturday, November 11, 2023
The University of Alabama

All sessions will be held in the Gordon Palmer (GP) Hall on the campus of the University of Alabama, Tuscaloosa.

9:30am	**Refreshments**	GP 205
10:00am	**Welcome Remark**	GP 208
10:10am	**Faculty Presentation:** Zoran Grujic (UAB), On persistence of spatial analyticity in the hyper-dissipative Navier-Stokes models	GP 208
10:50am	**Faculty Presentation:** Daniel P. Bossaller (UAH), List Decoding: What to Do When You've Made Too Many Mistake	GP 208
11:30am	**Graduate Student Presentation:** Yiming Ren (UA), FFT accelerated high order finite difference method for solving elliptic BVP and interface problem	GP 208
11:50am	**Lunch on your own**	Area Restaurants
1:40pm	**Faculty Presentation:** Tiziana Giorgi (UA), Dimensional Reduction for the Ferroelectric SmA-Type Phase of Bent-Core Liquid Crystals	GP 208
2:20pm	**Graduate Student Presentation:** Paige Gant (UAH), Zero Forcing on Graphs with Prescribed Girth	GP 208
2:40pm	**Graduate Student Presentation:** Dipak P Dulal (UAB), Forecasting Covid-19 Infections via Sampling Techniques for Low-Rank CP Decomposition	GP 208
3:00pm	**Refreshments**	GP 205
3:20pm	**Faculty discussion**	GP 208
3:20pm	**Student discussion**	GP 206

Parking permits are not required on this Saturday. You can park wherever space is available on campus.

Abstracts

On persistence of spatial analyticity in the hyper-dissipative Navier-Stokes models

Zoran Grujic (UAB)

(joint work with Aseel Farhat, Florida State University)

The goal of this presentation is to demonstrate that as soon as the hyper-diffusion exponent is greater than 1, a class of finite time blow-up scenarios consistent with the analytic structure of the flow (prior to the possible blow-up time) can be ruled out. The argument is self-contained, in spirit of the regularity theory of the hyper-dissipative Navier-Stokes system in 'turbulent scenario' developed by Grujic and Xu. Recall that according to the classical theory (since J.L. Lions in 1960s), in order to rule out a finite time blow-up, the hyper-diffusion exponent needs to be greater or equal to 5/4.

List Decoding: What to Do When You've Made Too Many Mistakes

Daniel P. Bossaller (UAH)

When any electronic information is sent between computers, it is inevitable that there will be some errors included in the message in transmission. Algebraic coding theory seeks to find ways to strategically include sufficient redundancy so that (after some work) the message is readable by the receiver. However, when too many mistakes occur in transmission, the message may be totally unreadable. When this occurs, list decoding algorithms seek to recover a short list of possible code words which includes the intended message. This talk will survey the theory of list decoding and some of my recent research with Daniel Herden (Baylor) which investigates the list decoding problem in skew Reed-Solomon codes.

FFT accelerated high order finite difference method for solving elliptic BVP and interface problem

Yiming Ren (UA)

In this talk, an augmented matched interface and boundary method is introduced for solving elliptic boundary value problems and interface problems. First, A new finite difference method is proposed based on the matched interface and boundary (MIB) and fast Fourier transform (FFT) schemes, which achieves a fourth-order convergence and O(NlogN) efficiency. The

method involves a ray-casting MIB scheme to handle different types of boundary conditions and an augmented MIB formulation for efficient inversion of the discrete Laplacian using the FFT algorithm. The accuracy and efficiency of the proposed method are numerically examined for various elliptic BVPs in two and three dimensions. Second, A FFT accelerated finite difference method is proposed for solving a three-dimensional elliptic interface problem involving a smooth material interface inside a cuboid domain. The method can handle complex geometry using a fourth-order ray-casting MIB scheme and Cartesian derivative jumps as auxiliary variables. The proposed method is efficient and accurate, achieving an overall efficiency on the order of O(n^3logn) for a n×n×n uniform grid.

Dimensional Reduction for the Ferroelectric SmA-Type Phase of Bent-Core Liquid Crystals
Tiziana Giorgi (UA)

We model the effects of a bias constant electric field on a thin sample of a bent-core liquid crystal in the ferroelectric smectic A-type phase by considering a small thickness limit of a three-dimensional phenomenological energy functional. We show that under proper rescaling the electric self-interactions give rise to boundary terms and present numerical simulations based on the reduced model. This is a joint work with Carlos Garcia-Cervera and Sookyung Joo.

Zero Forcing on Graphs with Prescribed Girth
Paige Gant (UAH)

The zero forcing number of a graph is the minimum size of a zero forcing set of vertices. It has many applications ranging from minimum rank problems and control theory to monitoring power grids and searching buildings. This has prompted research into the parameter in its own right. This talk will focus on research into finding lower bounds for the zero forcing number of graphs with prescribed girth and minimum degree. Particular attention will be directed towards finding optimal lower bounds for the zero forcing number of graphs with girth five and minimum degree five and graphs with girth six and minimum degree five.

Forecasting Covid-19 Infections via Sampling Techniques for Low-Rank CP Decomposition
Dipak P Dulal (UAB)

In this talk, we use the canonical polyadic (CP) decomposition to analyze Covid-19 data. The Covid-19 tensor data is constructed from the US Census 2020 and NY Times Covid-19 databases.

We use a tensor completion framework for model prediction of future Covid-19 infections. Specifically, future Covid-19 infection densities are predicted for up to a week with high accuracy. In addition, future Covid-19 infection patterns and trends can be predicted for up to one quarter (three months). We also propose a sampling technique to approximate CP decomposition that reduces computational complexity. In addition, we implement a regularized sparse tensor optimization to approximate the tensor rank and reconstruct a low rank tensor of a given tensor data with high accuracy. Finally, we compare the proposed methods with standard tensor decomposition algorithm with respect to their interpretability, predictability, pattern recognition, visualization, cost analysis and computational time to demonstrate our methods' efficacy.